

Can the variance after-effect distort stock returns?

Tony Berrada

Discussant: Boone Bowles

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The *TEMPERATURE* After-Effect

Texas was really hot in August.

Even though temperatures were normal in September, they *seemed cold* since August was so hot.

When updating my beliefs about the temperature for October, I am going to be biased toward predicting cold weather since September *felt* so cold.

The *VARIANCE* After-Effect

Texas Instruments was really volatile in August.

Even though volatility was normal in September, Texas Instruments *seemed less volatile* since August was so crazy.

When updating my beliefs about the true volatility of Texas Instruments, I am going to be biased toward predicting low variance since September *felt* so calm.

The *VARIANCE* After-Effect

Texas Instruments was really volatile in August.

Even though volatility was normal in September, Texas Instruments *seemed less volatile* since August was so crazy.

When updating my beliefs about the true volatility of Texas Instruments, I am going to be biased toward predicting low variance since September *felt* so calm.

If investors experience the variance after-effect, what implications does that have for trading and asset prices?

Modeling the Variance After-Effect

The VAE is modeled as an investor trying to learn about the variance of an asset, but the VAE influences their updating.

$$\beta_{t+1} = \beta_t + \frac{[\sum (x_i - \mu)^2]_t}{2}$$

by introducing an adjustment factor Θ_t

$$\beta_{t+1}^{\text{ae}} = \beta_t^{\text{ae}} + \Theta_t \frac{[\sum (x_i - \mu)^2]_t}{2} \quad (1)$$

where

$$\Theta_t = \left[1 + \left(\frac{n-3}{n-1} \frac{[\sum (x_i - \mu)^2]_t}{[\sum (x_i - \mu)^2]_{t-1}} - 1 \right) \lambda \right]. \quad (2)$$

Portfolio Choice

The investor constructs an optimal portfolio of the riskless asset and a risky asset, but has to learn about the variance of the risky asset while being influenced by VAE.

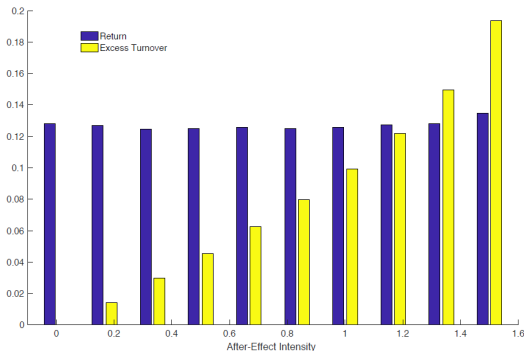


Figure 3: Portfolio turnover and performance based on 50000 simulations. The model coefficients are $\mu = 0.1$, $\sigma = 0.2$, $n = 20$, $T = 12$, $A = 2$. The variance after-effect intensity λ ranges from 0 to 1.5.

Financial Equilibrium Model

The investor trades a risky asset while learning about the variance and being influenced by the VAE.

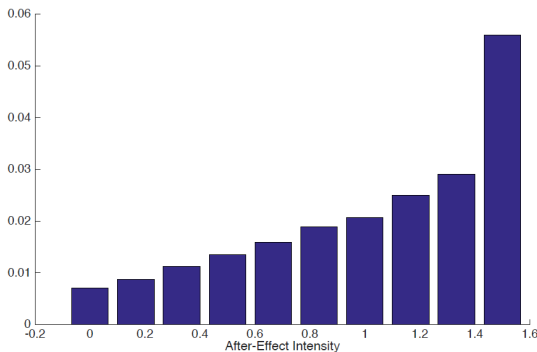


Figure 7: Equilibrium stock return volatility based on 10000 simulations. The model coefficients are $\mu = 0.2$, $\sigma = 0.1$, $n = 10$, $T = 20$, $A = 2$. The variance after-effect intensity λ ranges from 0 to 1.5.

Quick Note

Really cool paper!

An *approachable* theory paper.

It was exciting to read and has me still thinking...

“yes... and what else?”

A Few Points: Number 1

The experimental data suggests VAE intensity (λ) is typically ≤ 0.5 ...

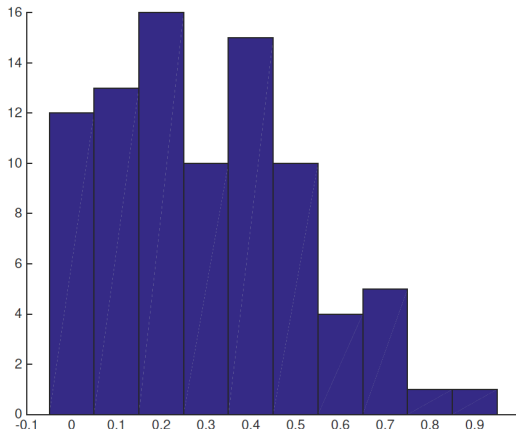


Figure 2: Histogram of the implied values of the variance after-effect intensity λ . The values are calibrated to match the experimental data of treatment 1 in [Payzan-LeNestour et al. \(2016\)](#).

A Few Points: Number 1

... but most of the *bite* of VAE happens when $\lambda \geq 1.0$

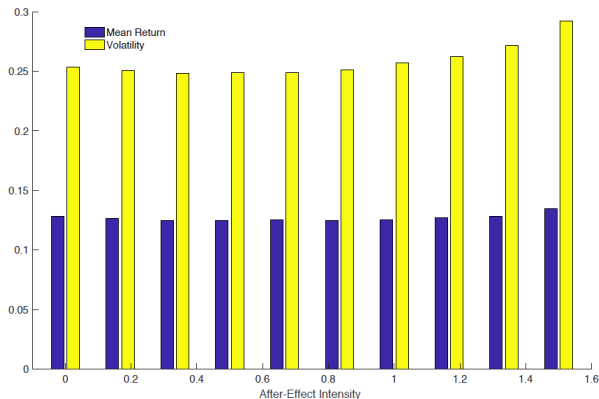


Figure 4: Portfolio turnover and performance based on 50000 simulations. The model coefficients are $\mu = 0.1$, $\sigma = 0.2$, $n = 20$, $T = 12$, $A = 2$. The variance after-effect intensity λ ranges from 0 to 1.5.

A Few Points: Number 1

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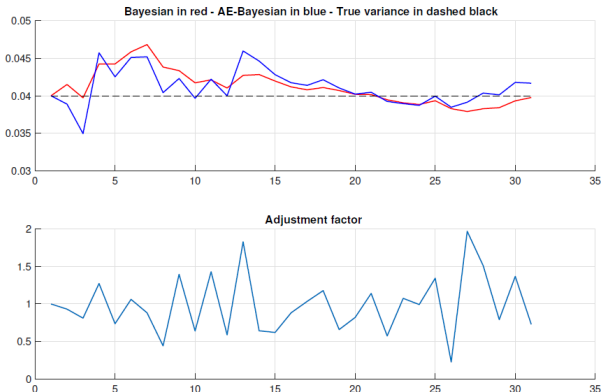


Figure 5: Sample path of variance estimates for bayesian (red) and AE-bayesian (blue), and adjustment factor. The model coefficients are $\lambda = 1$, $\mu = 0.1$, $\sigma = 0.2$, $n = 20$, $T = 30$, $A = 2$.

A Few Points: Number 1

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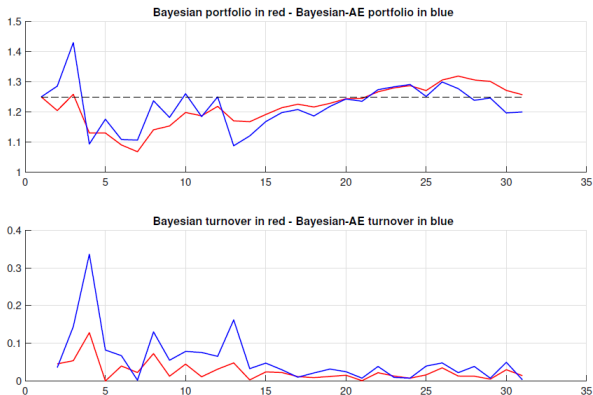


Figure 6: Sample path of portfolio and turnover for bayesian (red) and AE-bayesian (blue). The model coefficients are $\lambda = 1$, $\mu = 0.1$, $\sigma = 0.2$, $n = 20$, $T = 30$, $A = 2$.

A Few Points: Number 1

... but most of the *bite* of VAE happens when $\lambda \geq 1.0$

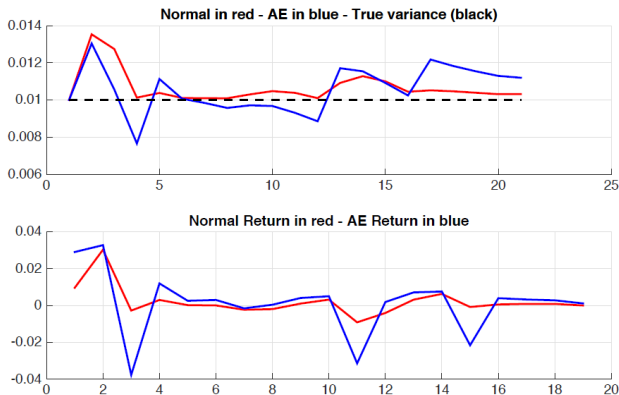


Figure 8: Equilibrium stock return and variance estimate. The model coefficients are $\mu = 0.2$, $\sigma = 0.1$, $n = 10$, $T = 20$, $A = 2$. The variance after-effect intensity is set at $\lambda = 1$.

A Few Points: Number 1

Provide some more understanding of λ and give us an idea of the λ that would be appropriate for financial professionals.

Is there a reason that the experimental λ of mostly $\leq .50$ is not good in your setting?

If $\lambda \leq .70$ is used throughout your simulations, do you have the same paper?

If you have reason to believe that $\lambda \geq 1.0$ other than Figure 2, please give it to us.

A Few Points: Number 2

VAE suggests that if in the distant past we have had high variance and then in the more recent past we have had normal variance, we perceive that variance to be low.

How does this fit with Recency Bias?

- Seems like they fit to me, but I would like to see some discussion on this.

How does this fit with Saliency Bias?

- Does the saliency bias work against VAE?

A Few Points: Number 3

Is there an asymmetry between HIGH past variance and LOW past variance?

This is only slightly addressed in the empirical section of the paper. I would like to see a lot more of this.

It could be the case that there is no asymmetry in the portfolio choice or financial equilibrium models, but if that's the case I couldn't see it clearly and instead expect the opposite.

Some Minor Points

- ① Tighten up the writing – make it easier on the reader.
 - It was often difficult to tell if you were talking about λ or Θ .
 - Is the updating procedure for hyperparameter α important? Maybe not as important as β , but I would like to know something about it.
 - I love the current brevity, but need just a bit more to help me see the theory.
- ② Does the VAE have to be driven by past variance of the same stock, or can high market variance or high competitor variance influence perceived variance of the subject stock?
- ③ Similarly, what about the Covariance After-Effect? This could enrich your portfolio choice model.
- ④ Asymmetric VAE for high versus low past variance.
- ⑤ **Costly learning (e.g., Kyle, Helwig, Grossman-Stiglitz) could be really interesting here.** How do the incentives to acquire costly information change when investors are influenced by the VAE. I'd love to chat further about this!